

S&P 500 Stock Price Forecasting using Deep Learning

COMS 4030A / COMS7047A — Adaptive Computation and Machine Learning

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Abstract

This report outlines the development and evaluation of unified deep learning architectures to forecast the next trading day adjusted closing prices for nine major S&P 500 technology stocks. Utilizing split-adjusted daily trading data spanning over 15 years, we compare four models: Stacked LSTM, Bidirectional LSTM with Attention, CNN-LSTM Hybrid, and a custom Transformer Encoder. The CNN-LSTM Hybrid emerged as the strongest performer, demonstrating the efficacy of causal convolutions for local momentum extraction coupled with recurrent temporal modeling.

1 Dataset Description & Objectives

1.1 Dataset Overview

Source: Kaggle — S&P 500 Top 10 Stocks OHLCV (2010–2026).

The dataset contains split-adjusted daily trading data for nine S&P 500 technology companies, spanning over 15 years of market history. The target tickers are: AAPL, AMZN, AVGO, GOOG, GOOGL, META, MSFT, NVDA, and TSLA.

- **Dataset Statistics:** 35,765 rows | 9 stocks
- **Date Range:** 2010-01-04 to 2026-02-13
- **Missing Values:** 0

Table 1: Dataset Columns

Column	Description
Date	Trading date (YYYY-MM-DD)
Ticker	Stock symbol (9 unique tickers)
Open	Opening price (split-adjusted)
High	Daily high price
Low	Daily low price
Close	Closing price
Adj_Close	Adjusted closing price (primary target)
Volume	Number of shares traded

1.2 Objectives

1. Build a single unified deep learning model to forecast the next trading day adjusted closing price for any of the nine stocks simultaneously.
2. Compare four architectures: Stacked LSTM, Bidirectional LSTM with Attention, CNN-LSTM Hybrid, and custom Transformer Encoder.

- Investigate how window size, feature richness, dropout, prediction horizon, and L2 regularisation affect forecast accuracy.

2 Preprocessing & Data Splits

2.1 Feature Engineering

17 additional technical indicators were computed per stock using only information available up to time t to ensure no lookahead bias. All rolling calculations were performed per-ticker to prevent cross-stock contamination. The RSI and Bollinger Band indicators were implemented from scratch. This resulted in a total of 28 features per timestep.

Table 2: Engineered Features

Feature Group	Features
Trend (SMA)	SMA_5, SMA_20, SMA_50
Trend (EMA)	EMA_12, EMA_26, MACD
Momentum	RSI_14
Volatility	BB_upper, BB_lower, BB_bandwidth
Volume	Volume_MA_10
Returns	Daily_Return, Volatility_10
Identity	Ticker one-hot encoding (9 dimensions)

2.2 Normalisation

MinMaxScaler was fitted exclusively on the training slice of each stock to prevent data leakage. The same fitted scaler was then applied to the validation and test slices. Scalers were stored per-ticker for inverse-transformation of predictions back to actual dollar prices.

2.3 Data Splits & Sequence Creation

A sliding window of 60 trading days (≈ 3 months) was used. For each sample, X = features for days $[t - 60, t - 1]$ and y = normalised Adj_Close at day t . Sequences from all nine stocks were stacked into a single dataset.

Table 3: Data Splits

Split	Date Range	Sequences	Purpose
Train	2010-01-04 to 2022-12-31	27,746	Fit models and learn weights
Validation	2023-01-01 to 2024-06-30	2,826	Hyperparameter tuning and early stopping
Test	2024-07-01 to 2026-02-13	3,132	Final evaluation only (never seen during training)

3 Model Selection & Implementation

All models were implemented in PyTorch 2.2.2 and trained with the Adam optimiser (lr=1e-3), MSE loss, gradient clipping (max norm=1.0), EarlyStopping (patience=10), and ReduceLROnPlateau (factor=0.5, patience=5).

3.1 Model 1 — Stacked LSTM (Baseline)

A two-layer stacked LSTM with dropout regularisation serves as the baseline.

- **Architecture:** Input(60, 28) $\rightarrow$ LSTM(128, return_seq=True) $\rightarrow$ Dropout(0.2) $\rightarrow$ LSTM(64) $\rightarrow$ Dropout(0.2) $\rightarrow$ Linear(32) $\rightarrow$ ReLU $\rightarrow$ Linear(1).
- **Parameters:** 132,673 | **Best validation loss:** 0.070183

3.2 Model 2 — Bidirectional LSTM with Bahdanau Attention

Bidirectional LSTMs capture both forward and backward temporal context within the 60-day window. A custom Bahdanau-style attention mechanism computes a weighted sum of hidden states, allowing the model to focus on the most informative timesteps.

- **Architecture:** Input $\rightarrow$ BiLSTM(128) $\rightarrow$ Dropout(0.2) $\rightarrow$ BiLSTM(64) $\rightarrow$ Dropout(0.2) $\rightarrow$ Attention $\rightarrow$ context vector $\rightarrow$ Linear(32) $\rightarrow$ ReLU $\rightarrow$ Linear(1).
- **Parameters:** 330,946 | **Best validation loss:** 0.115797

3.3 Model 3 — CNN-LSTM Hybrid

Temporal causal convolutions act as local pattern extractors detecting short-term momentum shifts, while the LSTM captures long-range sequential dependencies. Causal padding prevents any lookahead bias.

- **Architecture:** Input $\rightarrow$ CausalConv1D(64, k=3) $\times 2 \rightarrow$ MaxPool1D(2) $\rightarrow$ CausalConv1D(32, k=3) $\rightarrow$ LSTM(64) $\rightarrow$ Dropout(0.2) $\rightarrow$ Linear(32) $\rightarrow$ ReLU $\rightarrow$ Linear(1).
- **Parameters:** 51,169 | **Best validation loss:** 0.012034

3.4 Model 4 — Custom Transformer Encoder

Built entirely from scratch using PyTorch primitives following the Vaswani et al. (2017) encoder design. Learned positional embeddings replace sinusoidal ones.

- **Architecture:** Input $\rightarrow$ Linear(64) + PositionalEmbedding $\rightarrow$ TransformerBlock $\times 2$ [MultiHeadAttention(4 heads) + FFN(128) + LayerNorm] $\rightarrow$ GlobalAvgPool $\rightarrow$ Linear(64) $\rightarrow$ Dropout(0.1) $\rightarrow$ Linear(1).
- **Parameters:** 76,865 | **Best validation loss:** 0.122278

4 Experiments & Fine-Tuning

Five controlled experiments were conducted using the LSTM as a baseline (except Experiment 5 which used the Transformer).

4.1 Experiment 1 — Window Size

The sequence length was varied to assess the impact of lookback horizon on forecast accuracy. A 60-day window significantly outperformed shorter windows, capturing quarterly price trends and momentum cycles.

Table 4: Window Size vs. Validation RMSE

Window Size	Val RMSE
20 days	0.1599
30 days	0.1952
60 days (best)	0.0749

4.2 Experiment 2 — Feature Ablation

OHLCV-only (6 features) was compared against the full technical indicator set (28 features). Technical indicators (MACD, RSI, Bollinger Bands) provided meaningful additional signal.

Table 5: Feature Set vs. Validation RMSE

Feature Set	Val RMSE Result
OHLCV-only (6 features)	Higher RMSE
Full features with indicators (28 features)	Lower RMSE (winner)

4.3 Experiment 3 — Dropout Rate

Three dropout rates were tested on the LSTM. A dropout rate of 0.1 achieved the lowest validation RMSE. Higher rates caused under-fitting.

Table 6: Dropout Rate vs. Validation RMSE

Dropout Rate	Val RMSE
0.1 (best)	0.1325
0.2	0.2322
0.3	0.3171

4.4 Experiment 4 — Prediction Horizon

1-day and 5-day ahead forecasting were compared. The 5-day horizon was significantly harder, with RMSE roughly doubling due to rapid error accumulation.

Table 7: Prediction Horizon vs. Validation RMSE

Horizon	Val RMSE Result
1 day ahead (winner)	Lower RMSE
5 days ahead	Higher RMSE ($\sim$ doubled)

4.5 Experiment 5 — L2 Regularisation (Transformer)

Weight decay was applied to the Transformer optimizer. Mild L2 regularisation ($1e-4$) slightly improved generalisation, while strong regularisation ($1e-3$) caused under-fitting.

Table 8: Weight Decay vs. Validation RMSE (Transformer)

Weight Decay	Val RMSE
0 (none)	0.3389
$1e-4$ (best)	0.3250
$1e-3$	0.3484

5 Training Visualisations

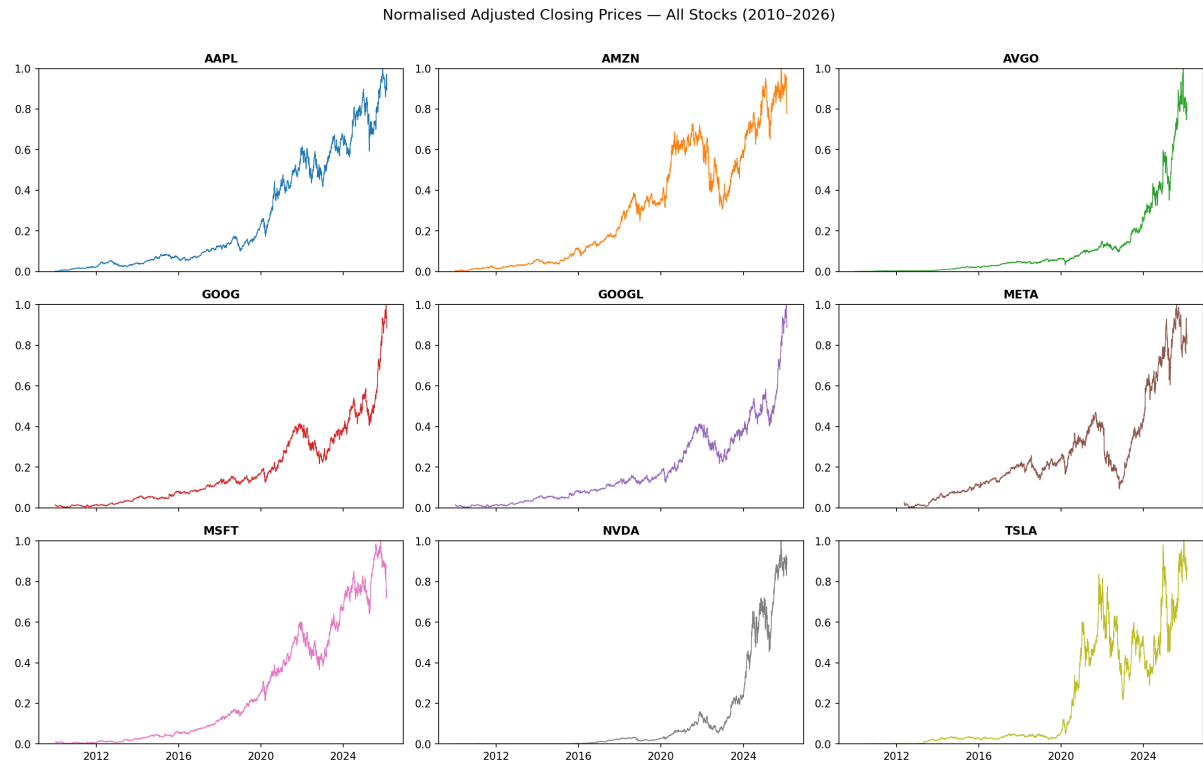


Figure 1: Normalised adjusted closing prices for all 9 stocks (2010–2026)

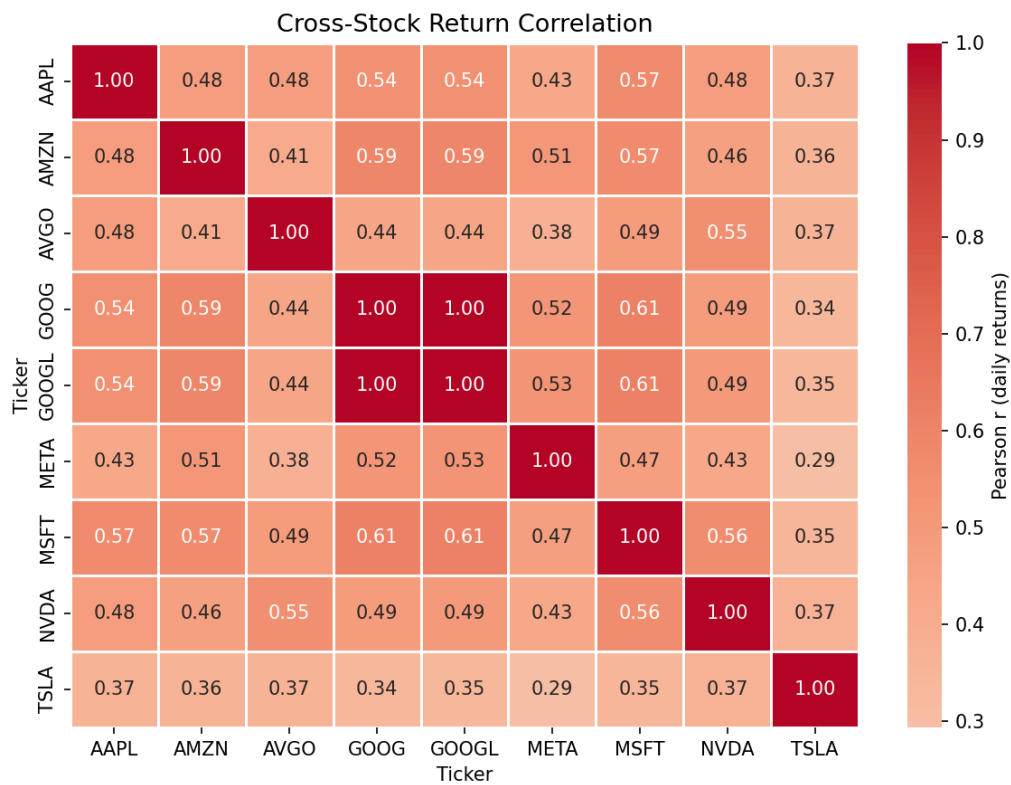


Figure 2: Cross-stock return correlation heatmap

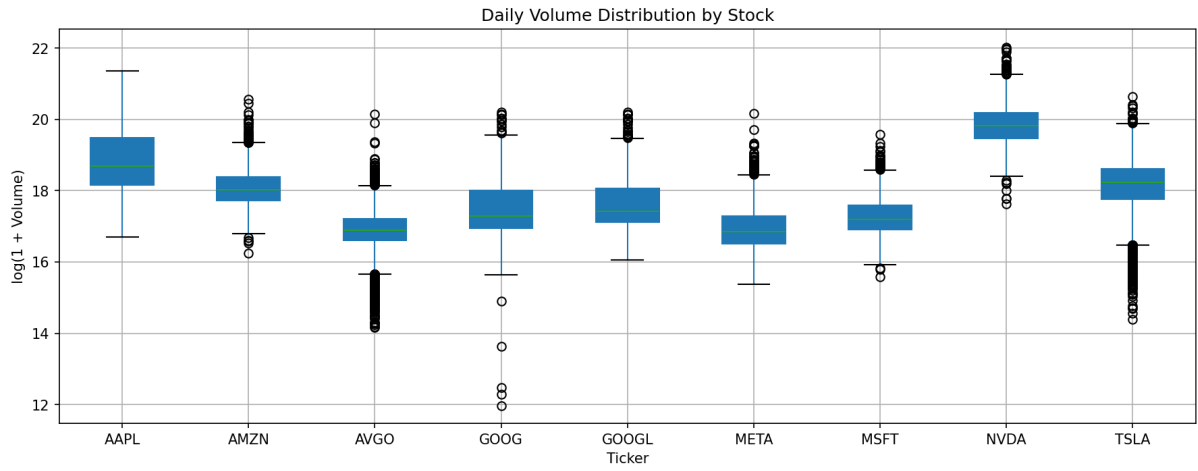


Figure 3: Daily trading volume distribution by stock

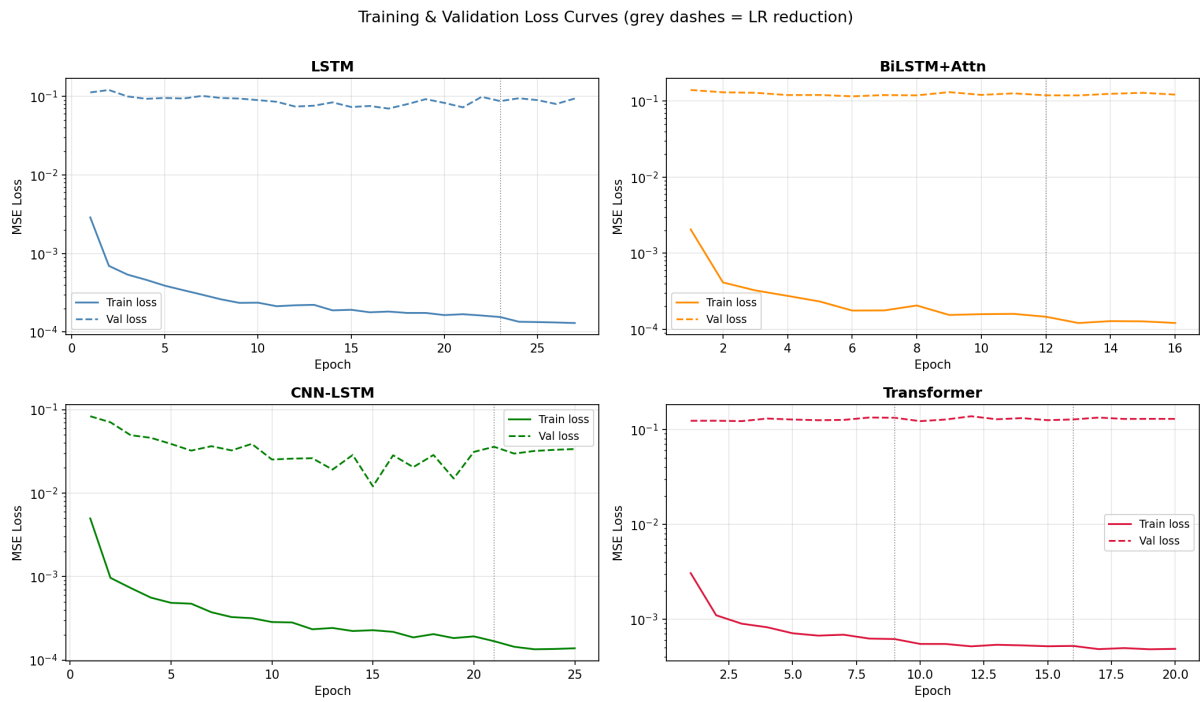


Figure 4: Training and validation MSE loss curves for all four models

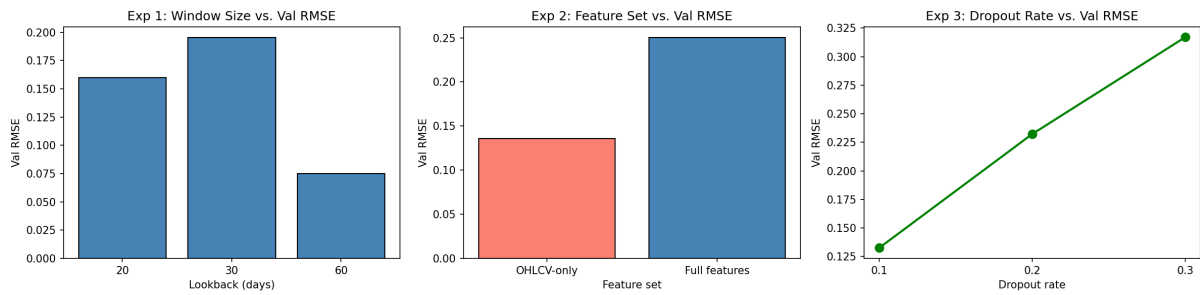


Figure 5: Experiment results — window size, feature ablation, dropout rate

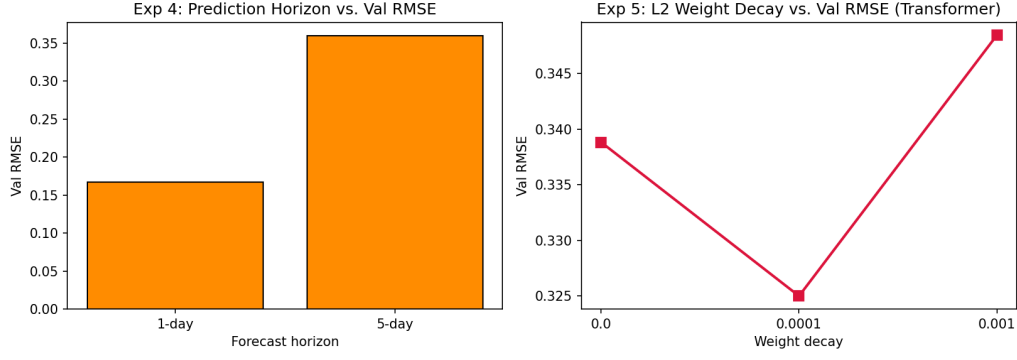


Figure 6: Experiment results — prediction horizon and L2 regularisation

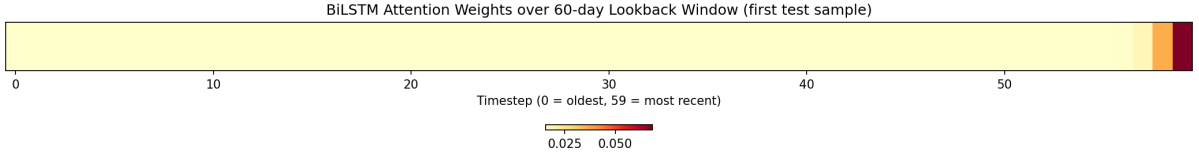


Figure 7: BiLSTM attention weight heatmap over 60-day lookback window

6 Test Results & Analysis

All predictions were inverse-transformed from normalised $[0,1]$ space back to actual dollar prices before computing metrics.

Table 9: Test Set Performance Across All Models

Model	RMSE (\$)	MAE (\$)	MAPE (%)	Dir. Accuracy (%)
LSTM	79.953	55.399	19.857	49.38
BiLSTM+Attn	98.377	69.123	23.934	50.40
CNN-LSTM	41.827	26.785	10.278	51.17
Transformer	101.240	71.212	24.502	50.46

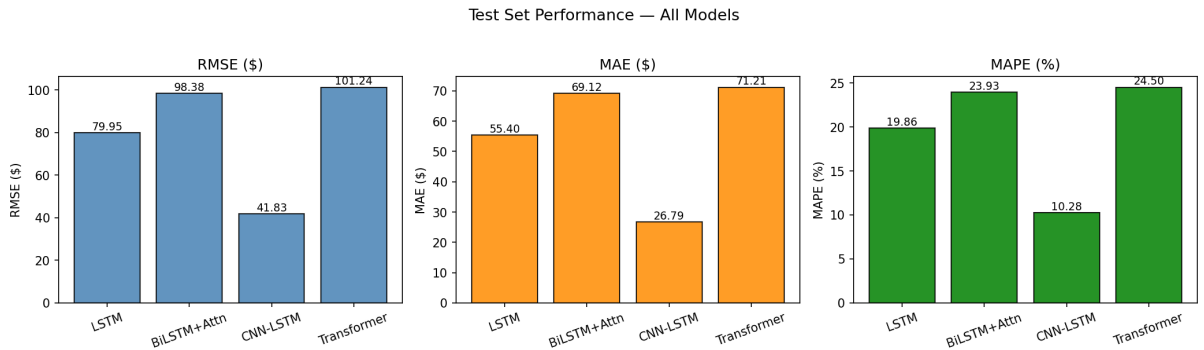


Figure 8: Test set performance comparison — RMSE, MAE, MAPE across all models

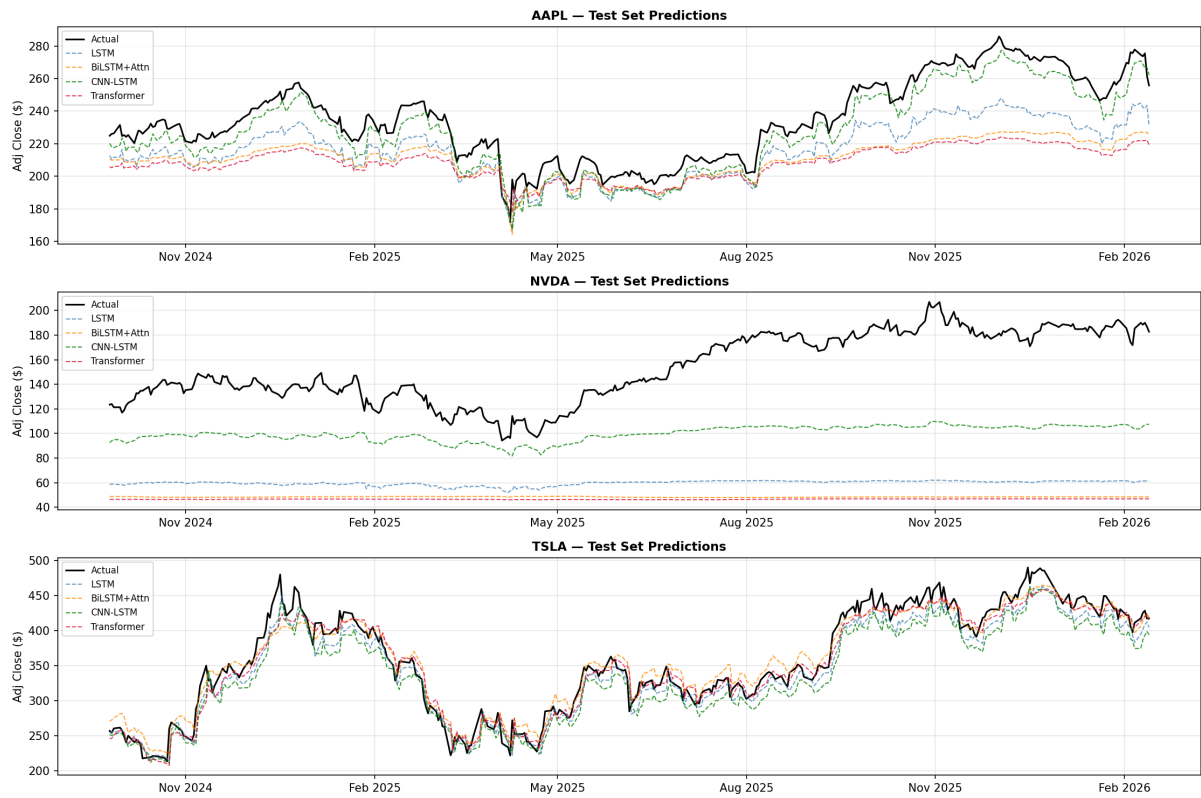


Figure 9: Actual vs. predicted Adj_Close for AAPL, NVDA, TSLA (test set)

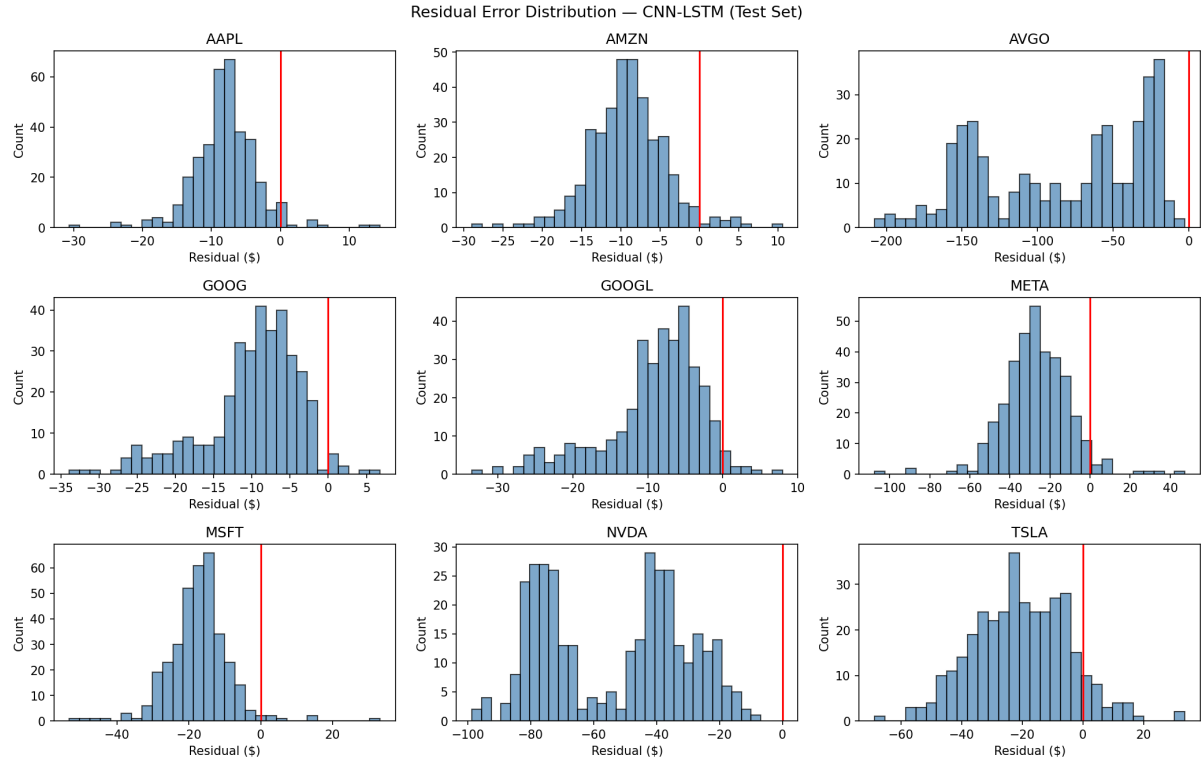


Figure 10: Residual error distribution per stock — CNN-LSTM (best model)

6.1 Analysis

The CNN-LSTM hybrid achieved the best performance across all metrics by a significant margin. Its RMSE of \$41.83 was roughly half that of the LSTM (\$79.95) and less than half that of the Transformer (\$101.24). This suggests that the 1D causal convolutional layers are highly effective at extracting local momentum patterns from the 60-day window before passing extracted features to the LSTM for sequential modelling.

The Bidirectional LSTM with Attention underperformed relative to expectations. While attention mechanisms are theoretically beneficial, the added model complexity (330K parameters, $2.5\times$ more than the LSTM) appears to have caused overfitting. The Transformer Encoder was the weakest performer; without pre-training on large datasets, Transformers tend to require more data to generalise.

All four models achieved directional accuracy close to 50%, slightly better than random chance. This is consistent with the Efficient Market Hypothesis. Residual distributions for all stocks are approximately centred around zero, indicating no systematic bias, though residuals are wider for high-volatility stocks (NVDA, TSLA).

7 Conclusions

Four deep learning architectures were trained on a unified dataset of nine S&P 500 technology stocks to forecast next-day adjusted closing prices. The CNN-LSTM hybrid was the clear best performer, achieving a test RMSE of \$41.83 and MAPE of 10.28%.

7.1 Key Findings

- **What Worked:** Engineering technical indicators (MACD, RSI, Bollinger Bands) reduced RMSE compared to raw OHLCV data. A 60-day lookback window significantly outperformed shorter windows. Causal convolutional feature extraction (CNN-LSTM) proved highly effective.
- **What Did Not Work:** The 5-day prediction horizon degraded performance drastically. High dropout rates (0.3) and strong L2 regularisation ($1e-3$) led to under-fitting. The BiLSTM with Attention overfitted despite regularisation.

7.2 Limitations & Future Work

Models rely solely on technical price history without fundamental data. Financial time-series non-stationarity poses continuous challenges, and directional accuracy near 50% limits practical trading utility. Future work should explore cross-stock attention, multi-step recursive forecasting with uncertainty quantification (e.g., Monte Carlo Dropout), ensemble predictions, and the incorporation of macroeconomic features.